

Part A — Voice Quality

Metric	Value	Target	Status
First-response latency	< 1.8s (avg)	< 2.0s	PASS
Transcription accuracy	~97% WER (Deepgram Nova-2)	> 95%	PASS
Booking completion rate	5/5 test calls (100%)	> 80%	PASS
Barge-in recovery	Handled via Vapi VAD	Required	PASS
Avg call duration	~3.5 min	—	INFO

Latency methodology: Measured from Vapi turn-end event to first audio byte from ElevenLabs, sampled across 10 test calls. Transcription accuracy estimated via reference transcript comparison.

Part B — Chat Groundedness

Hallucination Rate	5.6%	1/18 questions
Avg Groundedness	8.2/10	GPT-4o judge, 1-10 scale
Avg Coverage	7.9/10	Key concept coverage
Adversarial Pass Rate	100.0%	Injection/OOD rejection
Avg Chat Latency	1.84s	P95: 2.91s

Measurement method: 18-question golden Q&A; set (13 normal + 5 adversarial) evaluated by GPT-4o as judge model. Hallucination defined as: answer contains specific facts not present in retrieved context (repos, resume). Retrieval precision measured as fraction of retrieved chunks with score > 0.30 that contain answer-relevant text (manual verification).

Part C — Failure Modes Discovered

#	Failure Mode	Root Cause	Fix Applied
1	Gemini occasionally added facts not in retrieved context (soft hallucination)	Model temperature 0.7 too high; context window not emphasised strongly enough	Lowered temperature to 0.3; reinforced 'ONLY from context' instruction; added injection-detection layer
2	Voice agent gave 5-sentence responses on phone calls — too long	Same system prompt used for chat and voice without voice-specific constraints	Created separate VOICE_SYSTEM_PROMPT with hard 2-3 sentence limit; tested with 10 synthetic calls
3	Cal.com slots API returned empty list during late-night testing	Default availability window only covered 9am-6pm in owner's timezone; bug in dateFrom parameter	Extended days_ahead from 3 to 7; fixed UTC timezone handling; added mock_slots fallback for dev

Conscious Tradeoff Made

Accuracy vs Latency: Using OpenAI text-embedding-3-small for retrieval adds ~200-300ms per query vs a local embedding model (~30ms). I chose the hosted model because (a) its semantic quality is significantly higher — empirically fewer irrelevant chunks retrieved — and (b) for a demo/screening context, retrieval correctness matters more than shaving 200ms. In a production system with >1000 daily queries, I would cache frequent embeddings or move to a local model (e.g. bge-small-en) to reduce cost and latency simultaneously.

What I Would Build With 2 More Weeks

- 1. Streaming chat responses (SSE) to cut perceived latency from ~2s to sub-200ms first token
- 2. Hybrid retrieval: BM25 + dense vector search with Reciprocal Rank Fusion — higher recall for exact keyword queries (specific repo names, dates)
- 3. Automated eval regression suite in CI: run golden_qa.json on every git push, fail if hallucination rate > 5% or groundedness drops below 7.0
- 4. Voice call analytics dashboard: real-time transcript viewer, per-call scores, booking funnel visualisation
- 5. Multi-turn memory for voice: persist conversation state across tool calls so the agent doesn't re-ask for name/email

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